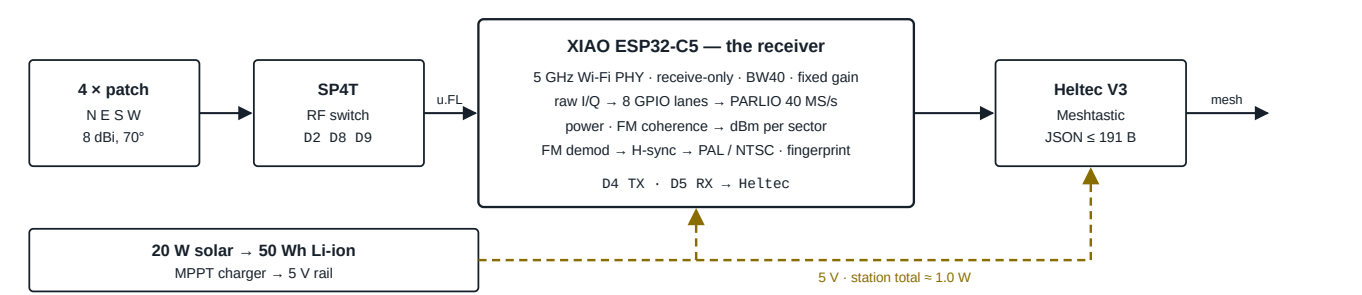


Level 1 Station v3 — C5 PHY prototype

The XIAO ESP32-C5's own 5 GHz Wi-Fi radio as the analog FPV receiver: sector direction finding and a software video check, with no RX5808 and no sync separator. Bench build and verification guide.

Document	docs/LEVEL1-V3-C5PHY-HARDWARE.md	Firmware	level1-c5phy/
Branch	level1-c5phy	Date	2026-09-22
Status	Firmware builds and links — nothing bench validated	Station	_____

SIGNAL CHAIN AT A GLANCE



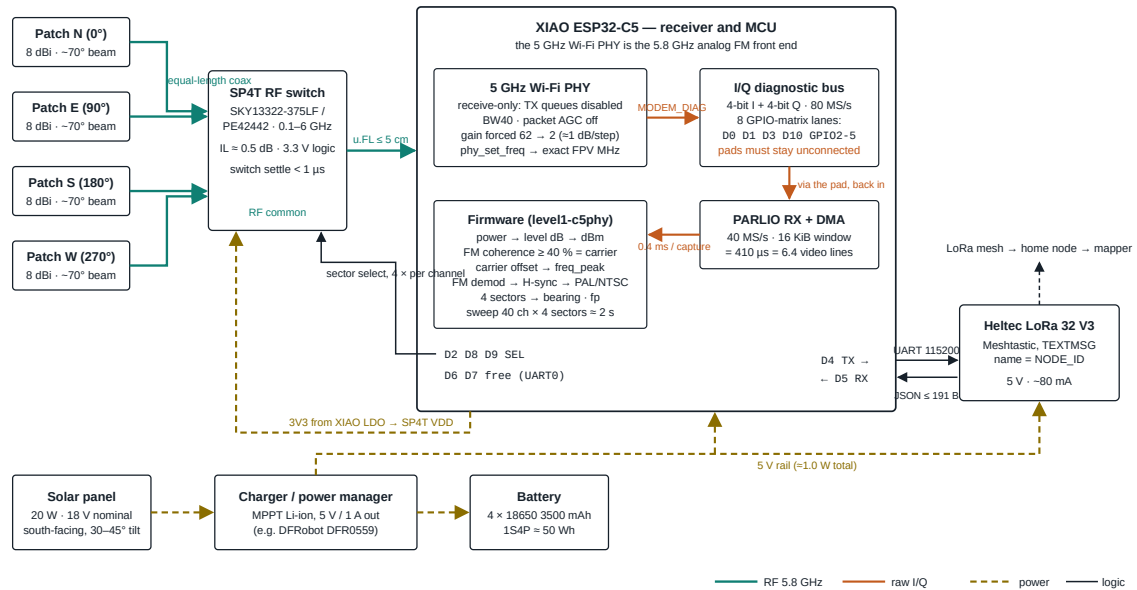
Simplified chain. The full block diagram with pin numbers, the I/Q path inside the C5 and the power tree is on sheet 2.

PARTS, AT A GLANCE

QTY	ITEM	KEY SPEC / PART	HAVE	≈ COST
1	XIAO ESP32-C5	receiver and MCU; only its U.FL antenna port is used	☐	\$10
4	Patch antenna	5.8 GHz, 8 dBi, ~70°, SMA	☐	\$32
1	SP4T RF switch	SKY13322-375LF / PE42442	☐	\$3
4	u.FL–SMA pigtail	equal length , ≤ 15 cm	☐	\$8
1	u.FL–u.FL pigtail	switch common → XIAO, ≤ 5 cm	☐	\$2
1	Heltec LoRa 32 V3	Meshtastic, TEXTMSG 115200	☐	\$20
1	Solar panel	20 W, 18 V nominal	☐	\$25
1	Charger / power manager	MPPT, 5 V / 1 A out	☐	\$15
4	18650 Li-ion	3500 mAh, 1S4P ≈ 50 Wh	☐	\$24
1	IP65 enclosure + brackets	≈200×150×100 mm, 10–15° wedge	☐	\$28
—	Passives	2 kΩ ×3, 100 nF ×2, 1000 μF	☐	\$2
—	5.8 GHz VTX + camera (bench only)	NTSC and PAL cameras; a step attenuator	☐	—
0	Not needed any more	RX5808, LMH1980/LM1881 sync separator, video coupling parts, RX5808 pigtail	—	–\$13

Before you start: nothing in this station has run on hardware yet. The receiver is the C5's Wi-Fi PHY driven through undocumented calls that the C5VRX project proved on ESP-IDF 6.0; this firmware re-implements them on the fleet's Arduino/ESP-IDF 5.5 toolchain. The procedure on sheets 6–8 is ordered so that each stage proves the next one is worth doing: receiver alive, channel coverage, selectivity, video (NTSC then PAL), sensitivity against an RX5808 station, then the switch and the pattern. Two rules that are easy to break: the eight I/Q lane pads (D0, D1, D3, D10 and the GPIO2–5 back pads) must stay unconnected, and every flash starts with a full erase. Full BOM on sheet 5.

1 · Full block diagram



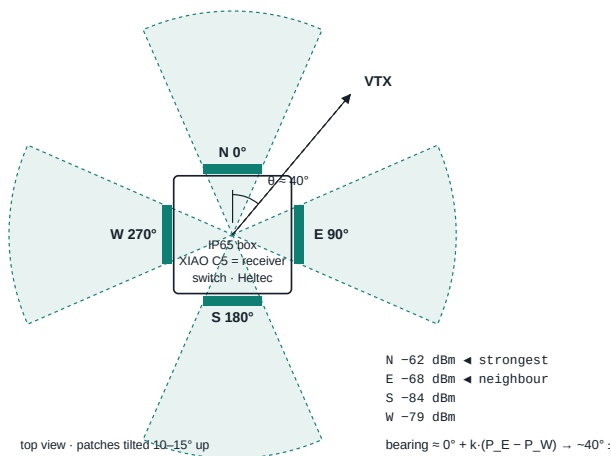
Four patches feed an SP4T switch whose common port goes straight into the XIAO's U.FL antenna connector. Inside the C5 the Wi-Fi PHY is held receive-only on the FPV frequency; its raw I/Q leaves on the modem's diagnostic bus, travels out through eight GPIO pads and back into the PARLIO peripheral, and the firmware measures power, FM coherence and horizontal sync on 16 KiB windows. The XIAO's 3V3 regulator feeds the switch; the 5 V rail feeds the XIAO and the Heltec.

WIRING LIST

FROM	TO	XIAO PIN	C5 GPIO	NOTES	✓
Switch V1	XIAO	D2	25	2 kΩ series	☐
Switch V2	XIAO	D8	8	2 kΩ series	☐
Switch V3	XIAO	D9	9	spare if the switch decodes 2 lines	☐
XIAO TX	Heltec RX	D4	23	UART 115200 — the same pins as every station tier	☐
XIAO RX	Heltec TX	D5	24	UART 115200	☐
Patches N/E/S/W	Switch RF1–4	—	—	equal-length coax ≤ 15 cm	☐
Switch common	XIAO U.FL	—	—	u.FL–u.FL ≤ 5 cm; the stock antenna comes off	☐
I/Q lane pads	nothing	D0 D1 D3 D10	1 0 7 10	must float: no resistor, no probe, no PCB trace	☐
I/Q back pads	nothing	GPI02 3 4 5	2 3 4 5	must float (underside JTAG pads)	☐
D6 / D7	free	D6 D7	11 12	UART0 (boot messages); free for a debug console	☐
5 V rail	XIAO 5V, Heltec 5V	—	—	1000 μF at the Heltec	☐
XIAO 3V3	Switch VDD	—	—	100 nF at the switch	☐
GND	all	—	—	single star point	☐

GPIO numbers are from the Arduino core's XIAO_ESP32C5 variant, used through `config.h`; wire by the D-labels on the silkscreen. The variant's D4 = GPIO23 / D5 = GPIO24 is now corroborated: C5VRX measured their resistor DAC on the XIAO's D4–D9 pads as GPIO 23, 24, 11, 12, 8, 9. The I/Q lanes are GPIO numbers (1, 0, 2, 7, 10, 5, 3, 4 in PARLIO data order), the set C5VRX proved on hardware. They need real pads: each MODEM_DIAG bit is driven out through the pad and read back by PARLIO from the same pad, so a load on the pad corrupts the samples.

2 · Sector layout and receiver



Top view. Each patch covers about 70°, so an emitter always sits inside one sector and near the edge of a neighbour. Unchanged from v2.

METHOD

Amplitude comparison. Take the strongest sector k and its two neighbours; with powers in dB the bearing is

$$\text{bearing} = \text{az}[k] + K \cdot (P[k+1] - P[k-1])$$

K , in degrees per dB, comes from the measured patch pattern. A bench sweep of a VTX around the box every 15° fills the lookup table. Report bearing_deg and bearing_sigma_deg; sigma grows when the two neighbours are nearly equal, or the peak is weak.

PER-CHANNEL SCAN LOOP

```
rf_tune(channel); delay(8 ms)
for sector in N,E,S,W:
    select(sector); wait 200 μs
    gain = 62
    for w in 1..3 (1 on a quiet channel):
        capture 16 KiB @ 40 MS/s      (0.4 ms)
        metrics                        (~2 ms)
        if clip > 3 %: gain -= 12; retake
    level = min(windows); q = median
    hit = level[best] >= 8 dB and q[best] >= 40 %
```

Sweep $\approx 40 \times (10 + 4 \times 8 \text{ ms}) \approx 1.7 \text{ s}$ at the full window count. The gain steps keep strong carriers unclipped and weak ones at full gain; the minimum over windows drops Wi-Fi bursts.

HOW THE C5 BECOMES THE RECEIVER

STEP	WHAT HAPPENS	WHERE
Bring-up	Wi-Fi driver in station mode with disconnected power management off; 5 GHz only, no power save, BW40; promiscuous with an empty filter; the five MAC transmit queues hardware-disabled; packet AGC off; PLL-tracking timer stopped. The radio can no longer transmit.	c5phy_rf.cpp rf_start
Tune	Park on the nearest public 5 GHz centre the regulatory table accepts (esp_wifi_set_channel), then phy_set_freq to the exact FPV MHz; re-arm the diagnostic stream, re-assert AGC-off / BW40 / gain. Band A is exactly Wi-Fi ch 149–173.	c5phy_rf.cpp rf_tune
Capture	PARLIO reads the 8 lanes at 40 MS/s (every second modem sample) into one 16 KiB DMA window: 410 μs, one byte per sample, I high nibble, Q low nibble.	iq_capture.cpp
Measure	Mean power; share of samples with a phase step within $\pm 45^\circ$ of the previous one (q_phase); clipping; mean phase step → carrier offset; envelope variance (noise or not); phase-step variance (video swing or bare CW).	demod.cpp iq_metrics
Level	level_db = (62 - gain) × 1 dB + 10 · log10(p_mean / 2.0); 2.0 is the noise power at gain 62 (C5VRX measurement). rssi_dbm = -95 + level_db until calibrated on sheet 7.	main.cpp measure_sector
Hit	Level ≥ 8 dB and coherence ≥ 40 % on the strongest sector. The coherence gate is the selectivity: a carrier 10 MHz off steps 90° per sample and never counts, so a neighbour inside the 40 MHz filter raises the level but is not a hit here.	main.cpp scan_channel
Video	On the strongest due peaks after the sweep: FM-demodulate 8 windows and look for horizontal sync repeating at the PAL or NTSC line period. Sheet 4.	demod.cpp video_window

3 · Video check and serial output

VIDEO CHECK

On the strongest due peaks after the sweep: hold the channel and the sector, demodulate 8 windows 5 ms apart (phase step between samples = instantaneous frequency), find sync-tip pulses 3.2–7.2 μ s wide below a threshold set from the window's own percentiles, and measure the spacing of consecutive pulses.

LINE PERIOD	LINE RATE	VERDICT	MEANING
64.0 μ s (2560 samples)	15 625 Hz	PAL	real analog video carrier
63.56 μ s (2542 samples)	15 734 Hz	NTSC	real analog video carrier
sync-shaped pulses, no repeating period	—	none	not video-shaped: noise, Wi-Fi, an unmodulated carrier
swing under 4 units/sample	—	none	no FM deviation on this channel

A window counts with ≥ 2 consecutive periods and a sync score ≥ 70 (width closeness 0–40 + period closeness 0–60); the carrier is video when ≥ 3 of 8 windows count. sync_q is the share of windows that counted. The **field rate is not measured** (a 410 μ s window never sees a vertical interval): field_hz is the nominal 50/60 of the detected standard. The fingerprint fp is `<video>/<sync_hz>/<freq_peak>`, e.g. NTSC/15736/5734, where freq_peak is the tuned frequency plus the measured carrier offset — no synthesizer sweep needed.

EXPECTED SERIAL OUTPUT

Boot line on USB at 115200. "rf":true is the PHY receive-only and the PARLIO reader running; false is followed by an {"info":"error",...} line naming the failed call. The bench console (?, h, s, g, v, x) prints {"info":"bench",...} lines with level_db, gain, q_phase, p_mean, clip, cfo_khz, mod, noise, phy_rssi, nf_dbm.

```
{"info":"c5phy v3 station ready","node_id":"RX01","receiver":"c5phy","hw":"v3","channels":40,"sectors":4,"heading":0,"threshold_dbm":-87.0,"threshold_level_db":8.0,"q_min":40,"peak_pick":1,"video":1,"bw40":1,"gain_max":62,"window_us":409,"rf":true}
```

Detection line — the v2 contract with receiver/hw changed and diagnostics added; the mesh copy is byte-for-byte the v2 format:

```
{"type":"analog_fm","receiver":"c5phy","hw":"v3","mac":"AF:00:16:6C:52:03","freq_mhz":5732,"band":"R","ch":3,"rssi_raw":903,"rssi":903,"rssi_dbm":-68.1,"rssi_min":890,"rssi_max":915,"rssi_n":3,"level_db":26.9,"gain":50,"q_phase":71,"cfo_khz":1840,"carrier":"fm","sectors":[-68.1,-74.4,-91.0,-85.2],"sector":0,"bearing_deg":32,"bearing_sigma_deg":15,"freq_peak":5734,"video":"NTSC","sync_hz":15736,"field_hz":60,"sync_q":88,"sync_score":91,"video_windows":8,"fp":"NTSC/15736/5734","basic_id":"5.8G-R3-5732MHz","node_id":"RX01","seq":42}
```

Heartbeat every 60 s on USB, 120 s on the mesh:

```
{"heartbeat":true,"node_id":"RX01","receiver":"c5phy","hw":"v3","scanning":true,"channels":40,"sectors":4,"heading":0,"threshold":1010,"threshold_dbm":-87.0,"video_seen":7,"gain_max":62,"bw40":1,"tune_fail":0,"cap_err":0,"sweeps":1830,"nf_dbm":-98,"temp_c":41.2,"uptime_s":3600,"seq":42}
```

rssi_raw, rssi, rssi_min, rssi_max are **synthesised** from rssi_dbm through the mapper's own RX5808 curve, so a mapper without rssi_dbm support still draws the right ring. The mapper on main needs no change for a v3 station. tune_fail counts channels the regulatory table refused, cap_err failed I/Q captures, nf_dbm is the PHY's own noise-floor estimate.

4 · Full bill of materials

Single-unit planning prices, quantities per station. Total ≈ \$170 all-in; ≈ \$80 electronics only (v2: ≈ \$190 / \$95).

RF FRONT END

#	PART	QTY	NOTES	✓	≈ \$
1	Seeed XIAO ESP32-C5	1	Receiver and MCU; only its U.FL port carries RF, the stock antenna is not used.	☐	10
2	5.8 GHz patch, 8 dBi, ~70°, SMA	4	Linear is polarisation-agnostic (~3 dB vs RHCP); RHCP +3 dB common case, ~20 dB on LHCP. One batch, never mixed.	☐	32
3	SP4T RF switch, 0.1–6 GHz, 3.3 V control	1	SKY13322-375LF or PE42442: ~0.5 dB loss, sub-μs. Check control lines and truth table; eval board first (~\$40).	☐	3
4	u.FL–SMA pigtails, equal length ≤ 15 cm	4	Matched length keeps sector losses matched — the bearing depends on it.	☐	8
5	u.FL–u.FL pigtail ≤ 5 cm	1	Switch common port to the XIAO's U.FL. Shortest run.	☐	2
6	SMA bulkhead feedthroughs	4	One per enclosure face.	☐	8
7	5.8 GHz band-pass filter (optional)	1	If a nearby 5 GHz AP overloads the front end (the PHY's own filter is 40 MHz wide).	☐	5

MCU, MESH, POWER, MECHANICAL

#	PART	QTY	NOTES	✓	≈ \$
8	Heltec WiFi LoRa 32 V3 + antenna	1	Meshtastic, TEXTMSG 115200, node name = NODE_ID; D4/D5 as on every tier.	☐	20
9	Solar panel 20 W, 18 V	1	South-facing, 30–45° tilt. 10 W covers summer only.	☐	25
10	MPPT charger, 5 V / 1 A out	1	DFRobot DFR0559 or Waveshare Solar Power Manager.	☐	15
11	18650 Li-ion 3500 mAh, 1S4P holder	4	≈ 50 Wh. Protected cells or a BMS board.	☐	24
12	1000 μF low-ESR cap at the Heltec	1	LoRa TX bursts.	☐	1
13	IP65 enclosure ≈ 200×150×100 mm	1	Patches on the four vertical faces at 0/90/180/270°, tilted 10–15° up.	☐	15
14	Cable gland, desiccant	1		☐	3
15	Pole bracket + four patch wedges	1 set	Face N to true north and record the heading; a mis-oriented box rotates every bearing.	☐	10
16	2 kΩ ×3, 100 nF ×2	—	Switch control series resistors and supply decoupling.	☐	1

REMOVED FROM THE V2 STATION

V2 PART	REPLACED BY
RX5808 receiver, SPI-modded	The C5's Wi-Fi PHY: public-centre bootstrap + phy_set_freq for tuning; raw I/Q power at a fixed gain for RSSI.
LMH1980 / LM1881 sync separator, 0.1 μF, 75 Ω	Software: FM discriminator on the same I/Q, H-sync search, PAL/NTSC from the line period.
RX5808 pigtail, 10 μF, SPI and sync wiring	Nothing. D0/D10 become I/Q lanes, D6/D7 free pins, D8/D9 carry the switch.

Carrier PCB rules. XIAO C5 footprint with D0, D1, D3, D10 and the underside GPIO2–5 pads **unrouted** (no test points); SP4T with four u.FL launches and a u.FL to the XIAO; control lines on D2/D8/D9 via 2 kΩ; UART header on D4/D5; decoupling. Nothing else. Make it after stage 6.

5 · Bench procedure — is the receiver alive?

Work top to bottom on a bare XIAO with the stock antenna on the U.FL; the switch and patches come in stage 6. Do not move to the next stage until the current one gives the stated result. Console commands are typed into the serial monitor and Enter-terminated: ? status, h R3/h 5732 hold, s 0–3 sector, g 30/g a gain, v video check, x resume.

STAGE 0 — FLASH AND BOOT

- ☐ pio run -e seed_xiao_esp32c5 --target erase, then --target upload. A **full erase precedes every flash**: stale PHY calibration data left C5VRX's receiver deaf after reflashes. Boot mode if it will not connect: hold BOOT, tap RESET, release BOOT.
- ☐ Monitor at 115200. **Expected:** {"info":"c5phy v3 station ready",..., "rf":true}. If "rf":false, copy the {"info":"error"} line here: _____
- ☐ ? twice, 60 s apart. **Expected:** captures climbing (hundreds per second while scanning), cap_err 0. Measured: captures/min _____ cap_err _____

STAGE 1 — RECEIVER ALIVE

- ☐ No VTX on. h A1. **Expected:** level_db 0 ± 2 , p_mean ≈ 2 , q_phase under 15, gain 62, clip 0. Measured: level_db _____ p_mean _____ q_phase _____
- ☐ Note the PHY's own no-signal figures: phy_rssi _____ dBm, nf_dbm _____ dBm. p_mean far from 2 means RF_NOISE_POWER is not this board's value — write it down for the calibration.
- ☐ Key a VTX on A1 (5865 MHz) a few metres away. **Expected:** level_db up by ≥ 30 dB, gain stepping below 62, q_phase ≥ 50 , mod 1, noise 0, cfo_khz within ± 2000 . Measured: level_db _____ gain _____ q_phase _____ cfo _____
- ☐ Move the VTX away or unkey it. **Expected:** the readings return to the no-signal values on the next line. level_db rising without q_phase rising means power but no FM coherence: an off-channel carrier (check cfo_khz) or scrambled lane order.

STAGE 2 — CHANNEL COVERAGE

- ☐ x to resume scanning; wait for the first heartbeat (60 s). **Expected:** tune_fail 0, channels 40. Measured: tune_fail _____
- ☐ If not 0: h each band's channels and note the ones refused ("tune refused by regulatory table"): _____. Set RF_COUNTRY_CC to the deployment country, reflash (erase first), repeat. Value used: _____
- ☐ h E4 (5645, lowest) and h E8 (5945, highest). **Expected:** both hold; ? shows the bootstrap wifi_ch 132 and 177. Measured: _____ / _____

STAGE 3 — SELECTIVITY

- ☐ VTX on R3 (5732), scanning. **Expected:** a detection line for R3 within two sweeps; B1 (5733) and F1 (5740) may hit too and PEAK_PICK folds them into one. Channels reported: _____
- ☐ **Expected:** no hit on any channel more than 40 MHz away. If one appears, h it and note q_phase _____ (a hit needs ≥ 40) and cfo_khz _____
- ☐ h R3 with the VTX on: cfo_khz _____ → the detection line's freq_peak _____ MHz. **Expected:** a stable value, the VTX's real offset.
- ☐ Hold the FPV channel nearest a live 5 GHz access point (A-band channels 5745–5825 overlap Wi-Fi 149–165). **Expected:** raised level_db but q_phase under 40 and no hit while scanning. Measured: level_db _____ q_phase _____

6 · Bench procedure — video and sensitivity

STAGE 4 — VIDEO CHECK

- ☐ VTX with an **NTSC** camera, hold its channel, v. **Expected:** ≥ 3 of 8 windows video:1, std:NTSC, line_hz 15734 $\pm$ 10, present:1. Measured: _____ / 8 windows, line_hz _____
- ☐ Same with a **PAL** camera. **Expected:** std:PAL, line_hz 15625 $\pm$ 10. Measured: _____ / 8, line_hz _____. **PAL is unproven on this receiver:** if it fails, record every window's swing, pulses, periods and period.
- ☐ VTX keyed with the camera unplugged. **Expected:** present:0; while scanning the detection line carries "video": "none" and carrier cw or fm.
- ☐ Back to scanning with the camera on. **Expected:** the detection line shows video, sync_hz and fp. Copy fp here: _____

STAGE 5 — SENSITIVITY AND DBM CALIBRATION

- ☐ VTX through a step attenuator (or a power meter on its output) into the U.FL; hold the channel and fill the level record below at each input level.
- ☐ **Expected:** level_db falls 1 dB per dB of attenuation over ~60 dB, gain rising back to 62 as the signal weakens, until the curve flattens at the floor. Measured slope: _____ dB/dB floor: _____ dBm
- ☐ Same VTX, same attenuator, on a v2 RX5808 station: floor _____ dBm. Difference v3 – v2: _____ dB. **This number decides whether v3 replaces v2 or supplements it.**
- ☐ Solve the two-point line and enter RSSI_CAL_DBM_AT_NOISE _____, RSSI_CAL_SLOPE _____, RSSI_CAL_OFFSET_DB (per station, antenna + cable) _____; reflash. Heartbeat threshold_dbm now reads _____ (expect -85 to -90).
- ☐ Quiet band, 60 s of h A1: highest level_db seen _____ → set DETECT_LEVEL_DB at least 6 dB above it: _____

LEVEL CALIBRATION RECORD

Known level into the U.FL on a held channel. The slope should be 1 dB/dB; the two-point line sets RSSI_CAL_DBM_AT_NOISE and RSSI_CAL_SLOPE in config.h. Where the firmware has stepped the gain down, level_db already includes the (62 – gain) term.

INPUT LEVEL (DBM)	-90	-80	-70	-60	-50	-40	-30
Measured level_db							
gain the firmware settled on							
q_phase (%)							
Hit while scanning? (y/n)							

No transmitter: level_db _____ p_mean _____ nf_dbm _____. VTX used: _____ power / attenuator: _____

Reading the numbers. level_db is dB above the noise reference (mean I^2+Q^2 of 2.0 at gain 62); rssi_dbm = RSSI_CAL_DBM_AT_NOISE + level_db × RSSI_CAL_SLOPE + RSSI_CAL_OFFSET_DB. The default -95 dBm is kTB over ~20 MHz plus a ~6 dB noise figure, not a measurement. A slope far from 1 dB/dB, or a floor much worse than the RX5808's, is a real result — record it, do not tune the constants to hide it.

7 · Bench procedure — switch, pattern, field, power

STAGE 6 — RF SWITCH AND PATTERN

- ☐ Eval board powered from 3V3, one patch on RF1, common port → XIAO U.FL, control lines from D2/D8/D9 through 2 kΩ. Drive each combination and record which RF port goes through. **Write the truth table here** (the firmware's V1 is D2):
V1V2V3 = 000 → RF__ 100 → RF__ 010 → RF__ 110 → RF__ → SECTOR_SWITCH_TABLE
- ☐ VTX on, hold its channel, s 0 ... s 3. **Expected:** level_db follows the sector the patch is on and settles within one line. Insertion loss against the patch wired direct to the U.FL: _____ dB (expect ≈ 0.5)
- ☐ Mount all four patches on the box, VTX at 30 m on a known bearing, clear line of sight. Rotate the box in 15° steps through a full turn, logging all four sectors values at each step.
- ☐ Derive K in degrees per dB from the neighbour difference. Value: _____ °/dB (default 3.0). Save the sweep to docs/ as the station's bearing calibration.

STAGE 7 — FIELD ACCEPTANCE

- ☐ Two v3 stations 300–500 m apart, both positions set in the mapper, both named to match their Meshtastic nodes, D4/D5 wired to the Heltecs.
- ☐ Walk a VTX between them on ten marked points. **Pass:** the 90 % circle contains the true position on at least 9 of 10. Score: _____ / 10
- ☐ **Pass:** video reads NTSC or PAL on every report for a live camera, never none; the mesh copy arrives with fp.
- ☐ Leave both stations on solar for 72 h. **Pass:** no brownout, heartbeat gap never exceeds 5 min. Measured station current at 5 V: _____ mA (expect ≈ 200)

POWER BUDGET

LOAD	CURRENT	POWER	NOTES
XIAO ESP32-C5, Wi-Fi PHY receiving continuously, 240 MHz	~120 mA @ 3.3 V	~0.45 W	via the 5 V rail and on-board LDO. The radio never sleeps and never transmits; PARLIO/DMA add little. Datasheet estimate — the stage 7 current measurement replaces it
SP4T switch	< 5 mA	0.02 W	the entire DF add-on
Heltec V3, Meshtastic, mostly RX	~80 mA @ 5 V	0.40 W	+ ~120 mA during LoRa TX bursts
Station total		≈ 0.9–1.0 W	≈ 23 Wh/day. The v2 solar sizing (20 W panel, 50 Wh battery, ~38 h of darkness at 1.3 W) carries this with margin. The RX5808's 0.6 W is gone and the radio's ~0.4 W takes its place

RECORD

Built by		Date	
Verified by		Date	
Station ID		True-north heading of face N	
Firmware commit		Platform / libphy checked	

Copy the numbers from every stage into docs/LEVEL1-V3-C5PHY-HARDWARE.md §7 (bench log). The v2 station's guide, docs/Level1-Station-v2-Bench-Guide.pdf, stays the reference for the switch truth table, pattern sweep and field acceptance details.